Research Note: Polymarket Post-Count Markets (Truth Social / Elon)

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Status: portfolio research write-up — **weakest empirical sample** of the Polymarket series
Stack: Python, XTracker counts, Polymarket Gamma + CLOB history, live signal toolkit
Code: this repository

Abstract

I built a regime / arithmetic framework for Polymarket **overlapping weekly post-count** markets (Trump Truth Social; exploratory Elon tweet weeklies). The useful idea is not "forecast the exact total," but translate each bin into a **required posts/day** on remaining time and fade upper bins that still need a sustained spike while the market prices them as central.

Honest headline: under the historically preferred live family (`NO` on upper bins only, ~24h after open, min edge 10%, one trade per window) the Trump backtest on **20** closed windows produces only **2** trades — both winners (**+0.385** PnL/share total). That is **interesting and operationally clean**, not a statistically proven income strategy. Broad `mode=all` takes **18** mostly-YES trades with **11%** win rate; total PnL stays positive only because of a **lottery skew** (one near-zero YES that paid ~+1).

Elon weekly support is **exploratory** (7 weekly windows matched; same tiny-N pattern).

The main result is sparsity, not a fat live edge



Validated Trump family (no_upper_only, 24h, edge≥10%, 1/window) — interesting, not statistically proven

Sample size

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1. Market structure

Series studied:

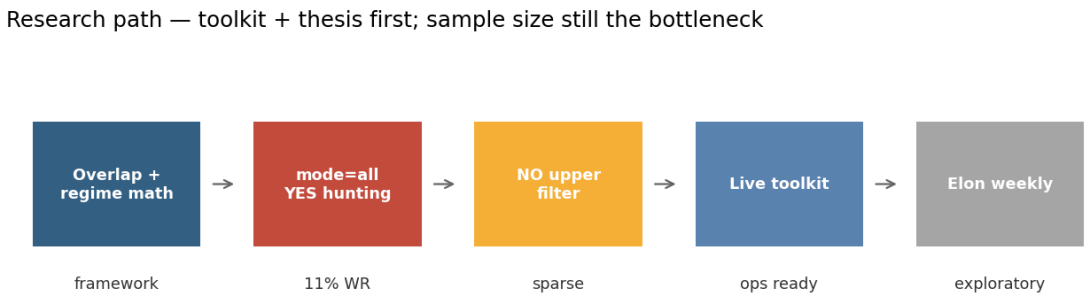
- Trump: `donald-trump-of-truth-social-posts-*` (Truth Social via XTracker)
- Elon: weekly `elon-tweets` windows (exploratory)

Windows **overlap**. Adjacent markets share day blocks, so prices that treat them as independent weekly totals are often wrong. Resolution follows XTracker-style counting (reposts / quotes matter; spikes move prices faster than fair value).

Core translation for bin `[L, H]`:

- `need_low = max(L - current_total, 0)`
- `req_low = need_low / days_remaining` (same for high)

Map required rate to crude regimes: quiet / base / active / spike. Trade when market-implied odds disagree with that regime map.



Research path

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2. What “validated” means here

In this repo, **validated** means: a trade family that survived historical filters and was wired into the live CLI defaults — **not** “large-N proven alpha.”

Selected Trump defaults (`scripts/live_signal.py`):

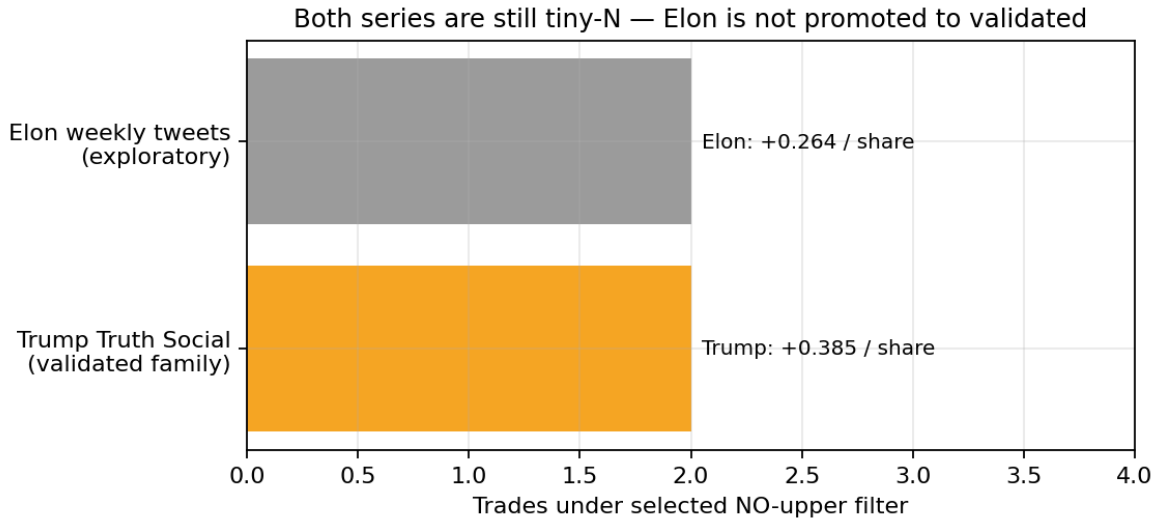
Knob	Value
Family	`no_upper_only` (NO on `80-99`, `100-119`)
Entry	first observation ≥ ~24h after window start
Min model edge	0.10
Cap	1 strongest trade per window

Backtest (20 closed windows, CLOB history + XTracker):

Metric	Value
Windows tested	**20**

Metric	Value
Trades	**2**
Wins	**2**
Win rate	**100%** (on n=2)
Total PnL / share	**+0.385**
Avg PnL / trade	**+0.193**

Both trades were `NO 100–119` in Sep/Oct 2026 windows. Treat 100% WR as a **red flag for sample size**, not a trophy.



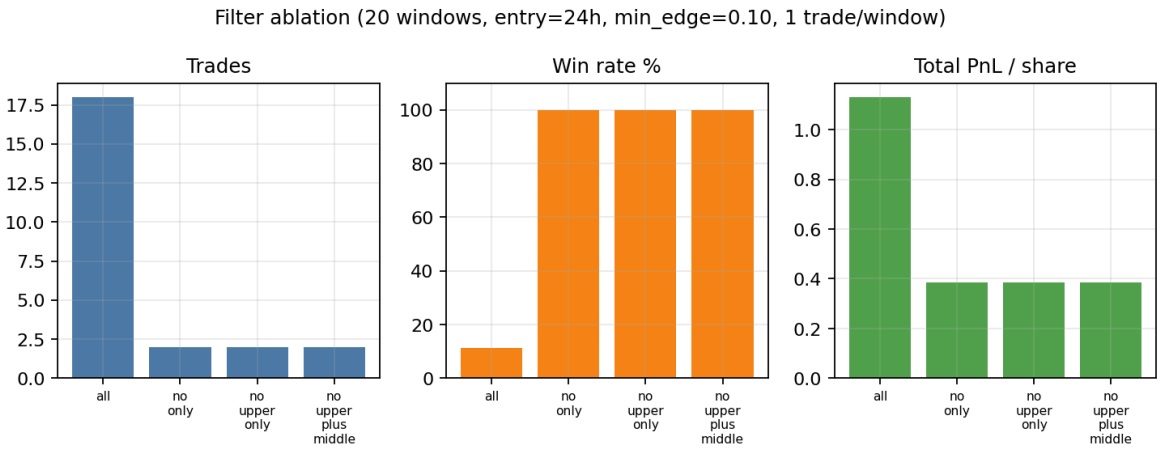
Series status

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3. Ablations — why the filter exists

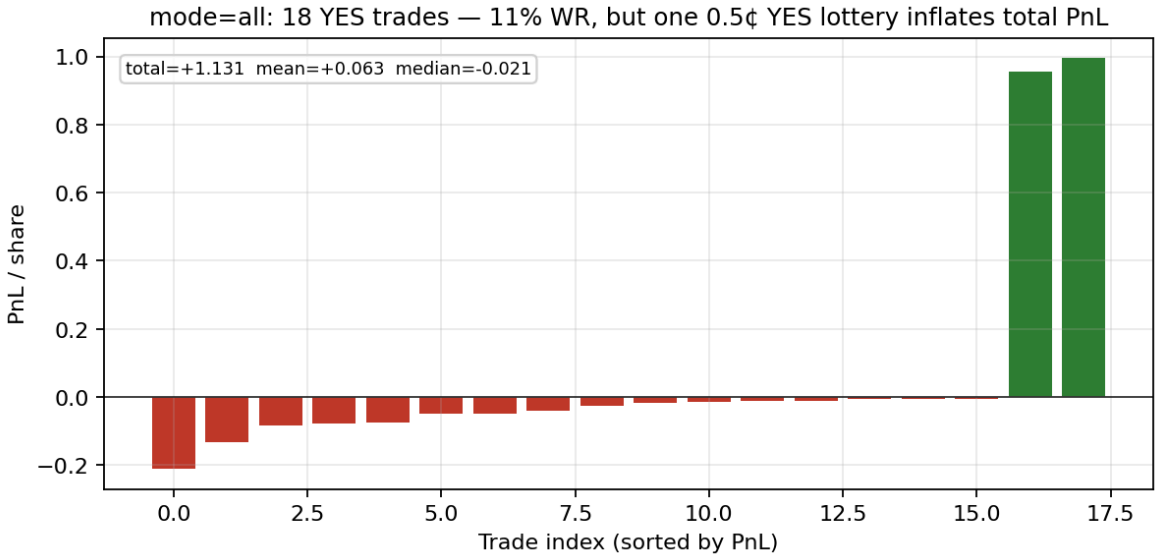
Same 20 windows, entry=24h, min_edge=0.10, max 1 trade/window:

Mode	Trades	Win rate	Total PnL / share
`all`	18	11.1%	+1.131
`no_only`	2	100%	+0.385
`no_upper_only` (selected)	**2**	**100%**	**+0.385**
`no_upper_plus_middle`	2	100%	+0.385



Mode comparison

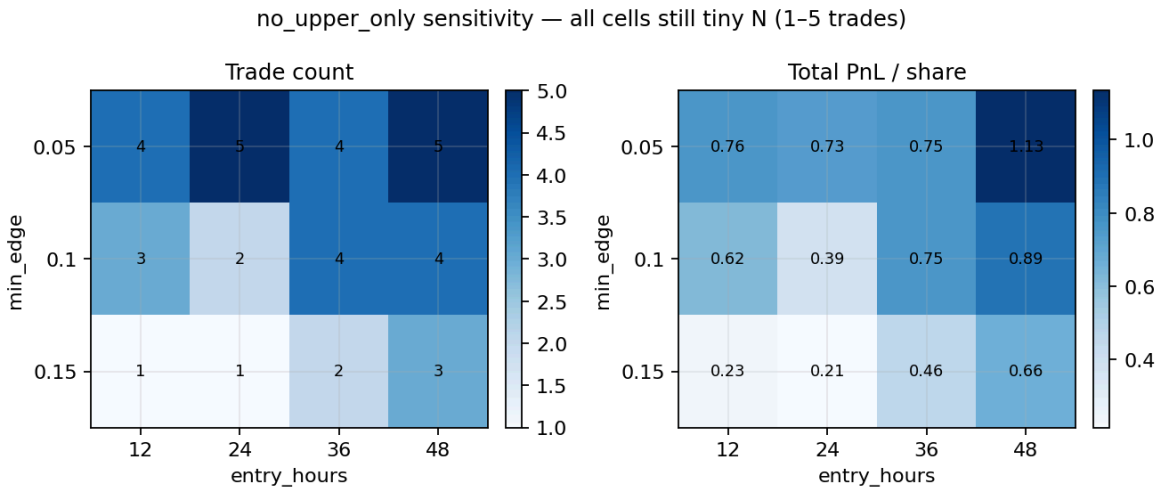
`mode=all` is almost entirely **YES**. Win rate collapses; total PnL is propped up by skew:



YES lottery

One YES entered near **0.5¢** and paid ~**+0.995**. Median trade in that set is still a small loss. **Do not** promote `all` as the live family.

Sensitivity on `no_upper_only` (entry hours × min edge) never escapes tiny N — cells are **1–5** trades, still often 100% WR:



Sensitivity

Loose edges (`min_edge=0.05`, entry 24–48h) add a few more trades and more total PnL, but the story remains **sparse selective NO**, not a high-throughput system.

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4. Elon weekly (exploratory)

`ELON_WEEKLY.md` / `scripts/elon_*` are intentionally **not** promoted to the same status as Trump.

Exploratory `no_upper_only` run (weekly windows only; 15 requested → **7** matched): **2** trades, **2** wins, **+0.264** PnL/share. Same pathology: filter fires rarely; perfect WR means almost nothing.

Next step before validation: longer sample, confirm a stable trade family, only then share live defaults with Trump.

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5. Ops toolkit (what is actually strong)

Even with weak statistics, the repo is a clean **research → signal → size → snapshot** loop:

1. `live_signal.py` / `daily_run.py` — current preferred Trump (+ Elon exploratory) printout
2. `position_size.py` — bankroll → simple share plans
3. `truth_social_rank.py` — full ranking with mode / window flags
4. `truth_social_snapshot.py` — JSONL state for later analysis
5. `backtest_truth_social_baseline.py` (+ sensitivity) — historical filter tests

That operational clarity is the portfolio asset. The missing asset is **sample size**.

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6. What this is / is not

Is: a curated note on overlapping post-count market microstructure, a selective `NO`-upper thesis, honest tiny-N backtests, and a usable daily toolkit.

Is not: a claim of durable live alpha; a high-frequency weather/BTC-scale evidence pack; or a finished Elon strategy.

Compared with sibling repos:

Repo	Evidence strength
polymarket-btc-5m-research	Strong negative-result economics story
polymarket-weather-research	Larger edge backtest + clear regime hierarchy
this repo	Strong framework, **weak N**

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7. Takeaways

1. **Overlapping windows + required-rate math** are the right abstraction.
2. **Unfiltered YES hunting looks profitable until you see the lottery skew.**
3. **Validated live defaults = sparse NO upper** — clean, not proven.
4. **n=2 at 100% WR is a warning label**, not a pitch.
5. Elon stays exploratory until the sample grows.

Reproduce

```
pip install -r requirements.txt
python scripts/make_figures.py

# network required — historical XTracker + CLOB
python scripts/backtest_truth_social_baseline.py \
  --windows 20 --entry-hours 24 --min-edge 0.10 \
  --max-trades-per-window 1 --mode no_upper_only
```

Figures rebuild from `results/` without re-hitting the APIs.